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# Enhanced Learning from Incidents Through Refinement of LOPC-Related HPI

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## Abstract

This work presents a structured and standardized approach to identifying High Potential Incidents (HPIs) arising from Loss of Primary Containment (LOPC) events within oil and gas industrial operations, with the goal of enhancing proactive learning and preventing Major Accident Events (MAEs). The current challenge in incident classification lies in the subjective interpretation of what constitutes a high potential scenario, particularly in complex environments such as offshore platforms and congested process areas. To address this, the comprehensive set of criteria has been developed by integrating in engineering calculations, barrier-based safety analysis, and incident management standards. The classification system is designed to be both technically robust and operationally simple, ensuring accurate identification of high-impact incidents while remaining accessible to field personnel.

The methodology introduces two complementary classification approaches:

1. **Immediate HPI Classification**, which includes Tier 1 and Tier 2 LOPC events, explosions, fires (including small, flash, and smoldering), and hydrocarbon gas detection in safe areas.
2. **Conditional HPI Classification**, which evaluates sensibly sized releases that may escalate under specific conditions. This involves predefined engineering calculations to assess release volumes and a review of existing safeguards to determine escalation potential. Furthermore, it integrates the evaluation of mitigation measures in place to collectively determine whether the event qualifies as a High Potential Incident (HPI).

   Gas release thresholds are based on simulations of cloud sizes capable of causing significant explosion damage, while liquid releases exceeding secondary containment capacity are flagged due to their potential to cause widespread fires. The presence or absence of effective mitigation barriers further informs the classification.

To ensure consistency and ease of use, the methodology is distilled into a simplified checklist and flowchart system. This approach enhances incident investigation protocols, strengthens corrective actions, and supports a proactive safety culture. By systematically extracting lessons from incidents with high potential consequences, the framework contributes novel insights to the field of process safety and aligns with the industry's goal of achieving zero incidents.

## Introduction

In the realm of process safety, particularly within the oil, gas, and chemical sectors, the accurate identification and classification of incidents with timely corrective action and preventing recurrence are paramount to preventing catastrophic events. Among these incidents, Loss of Primary Containment (LOPC) represents a critical failure mode that can lead to fires, explosions, and toxic releases. However, not all LOPC events carry the same level of risk, and distinguishing between routine incidents and those with high potential for escalation is essential for effective risk management.

Historically, the classification of High Potential Incidents (HPIs) has been influenced by individual judgment, leading to inconsistencies and potential underestimation of risk. Moreover, there is a lack of clear and standardized definitions or criteria for identifying HPIs arising from Loss of Primary Containment (LOPC) events in formal literature or industry standards, such as those provided by IOGP or API RP 754. Recognizing this gap, the project has been initiated to develop objective, criteria-based guidelines for determining HPIs resulting from LOPC events. This initiative aligns with broader efforts to enhance process safety through data- driven decision-making and standardized safety practices.

The development of HPI criteria draws upon multiple sources, including engineering design principles, incident management standards, and barrier-based safety models such as the Bow-Tie methodology. By integrating these elements, the proposed framework aims to provide a clear, replicable process for evaluating the potential severity of LOPC events and ensuring that high- risk scenarios are appropriately flagged and addressed.

This manuscript details the theoretical underpinnings, methodological approach, and practical application of the HPI criteria. It also presents the results of a pilot application involving LOPC cases, demonstrating the framework's utility in real incident cases.

Through this work, it seeks to contribute to the ongoing advancement of industrial safety practices and the prevention of major accident events.

## Statement of Theory and Definitions

Major Accident Events (MAEs) often originate from a Loss of Primary Containment (LOPC), which can lead to severe consequences such as fatalities, asset damage, reputational harm, and environmental impact. These consequences are driven by incident effects including fire impingement, heat radiation, overpressure, toxicity, etc. Among these factors, the release amount is the primary parameter determining the severity of the outcome.

In prevention of MAEs, the barrier concept is a safety philosophy that emphasizes the use of multiple, independent safeguards to prevent incidents or mitigate their consequences. Bow-Tie Analysis is a visual risk assessment tool that maps out the pathways from a hazard to an MAE, identifying preventive and mitigative barriers along the way.

### High Potential Incident (HPI)

A High Potential Incident (HPI) refers to an event that, while it may not have resulted in immediate harm or damage, possesses the credible potential to escalate into an incident with serious or crucial consequence, which includes Major Accident Event (MAE). HPIs are critical to identify and analyze because they serve as early warnings of systemic vulnerabilities within operational processes.

### Loss of Primary Containment (LOPC)

LOPC is defined as an unintentional release of hazardous substances from their intended containment systems, such as pipes, tanks, or vessels. These releases can be in the form of gases, liquids, or solids and may result from equipment failure, human error, or external factors.

**Major Accident Event (MAE)**

An MAE is a high-consequence incident that typically involves significant harm to people, assets, or the environment. Examples include explosions, large-scale fires, and toxic releases. Preventing MAEs is a central goal of process safety management.

**Tier Classification of LOPC**

LOPC events are categorized into tiers based on severity:

1. Tier 1: Significant release with substantial impact.
2. Tier 2: Moderate release with potential for escalation. These tiers help prioritize incident response and investigation efforts.

**Fire and Gas Detection Systems**

The theoretical foundation of FGDS lies in the concept of early hazard identification and automated risk mitigation. By detecting the presence of flammable gases, smoke, heat, or flames, these systems serve as mitigative barrier within the broader framework of process safety management. They are often integrated into Bow-Tie Analysis models as key safeguards against Major Accident Events (MAEs).

**Control of Ignition sources**

Ignition sources refer to any energy input capable of initiating combustion or explosion when in contact with a flammable atmosphere. These sources may be electrical (e.g., sparks, static discharge), mechanical (e.g., friction, hot surfaces), chemical (e.g., exothermic reactions), or thermal (e.g., open flames, heated equipment). Control measures include equipment design to meet hazardous area classifications (e.g., intrinsically safe or explosion-proof standards), grounding and bonding to prevent static discharge, hot work permitting systems, and continuous monitoring of surface temperatures. These controls are often embedded within barrier models such as Bow-Tie Analysis, serving as preventive safeguards against Major Accident Events (MAEs).

**Incident Management Standard (IMS)**

The standard establishes a structured process for incident reporting and analysis to ensure that all incidents are properly documented, investigated, and captured as lessons learned. It also defines the minimum requirements for the reporting, investigation, and follow-up of all incident types, including High Potential Incidents (HPIs), near misses, external complaints, non-compliance events, and others.

**IOGP**

The International Association of Oil & Gas Producers (IOGP) provides recommended practices for process safety including key performance indicators such as Tier 1 and Tier 2 classification for Loss of Primary Containment (LOPC) events, example of process safety incident, as well as guidance on Tier 3 and Tier 4 Indicators, etc.

## Description and Application of Processes

The HPI criteria for LOPC events are structured into two main parts, each addressing a different aspect of incident evaluation. These parts are derived from engineering calculations, safety standards, and empirical data.

**Part A: Immediate HPI Classification (Significant Breaching of Barriers)**

This step focuses on verifying whether a Loss of Primary Containment (LOPC) event could credibly escalate into a fire scenario, forming a critical part of High Potential Incident (HPI) assessment. Fire-related incidents are of particular concern in the oil and gas industry due to their potential for severe consequences, and this verification process helps ensure such risks are systematically identified and addressed.

Two primary aspects are considered in determining whether an event qualifies as a potential HPI.

**First**, any incident that results in an actual fire or explosion—regardless of size or severity—is treated as a potential HPI. This includes minor fires, flash fires, and smoldering fires. The rationale is based on the fact that fire and explosion hazards are typically controlled through multiple layers of preventive and mitigative safeguards, such as emergency shutdown systems, fire and gas detection systems, ignition control measures, and process alarms. The occurrence of a fire or explosion implies that several of these barriers have failed simultaneously. Such incidents represent a serious breakdown in risk control systems and are situated on the far-right side of the Bowtie diagram or the final layer in the Swiss cheese model, indicating a near-loss or loss of control over the hazard.

**Second**, incidents involving large volume releases that could potentially ignite and cause catastrophic damage are also considered. According to IOGP guidance, LOPC events categorized as Tier 1 or Tier 2 are indicative of multiple barrier failures. Tier 1 events involve significant consequences from actual loss of containment, while Tier 2 events, though of lesser consequence, may serve as precursors to more severe incidents.

In summary, the HPI criteria for part A include:

- Any explosion or fire event, regardless of containment or severity.
- Inclusion of minor fire types, such as small fires, flash fires and smoldering fires.
- Inclusion of Tier 1 and Tier 2 LOPC events, as defined by industry standards.

This systematic approach ensures consistent identification of HPIs and supports continuous improvement in process safety performance through the recognition and learning from high-risk scenarios, ensuring that even minor fire-related events are flagged for further analysis.

**Part B: Conditional HPI Classification**

This step focuses on identifying incidents involving significant releases that could have major impacts if their full potential consequences are realized. It begins with quantifying gas and liquid releases, combined with verifying mitigation safeguards — particularly for cases that fall outside the criteria defined in Part A. This is to assess whether the release, under specific conditions possesses credible potential to escalate into a Major Accident Event (MAE), thereby warranting further evaluation under conditional classification protocols.

***Gas/Liquid Release Evaluation.***

Gas Release Evaluation

The potential for fire or explosion from gas releases is assessed using literature, internal design guidelines and calculations:

- **Literature Review**: The HSE report went through a host of literature where flame speed and overpressure were measured in experimental trails. The trials covered a range of conditions, including methane and propane as the hydrocarbon source, and blockage rations ranging from 0-40%. The HSE report recommends a 5-meter grid for congested offshore platforms, based on experimental data showing that clouds smaller than 6 meters are unlikely to cause damaging overpressure.

  Large open facilities could have much larger clouds (10 meters or more) that will not result in significant damage because there is a lack of confinement and obstructions.

- **Internal Design Guidelines**: The fire and gas detection design guideline recommends about fire and gas detection mapping that gas detection objectives are set for 5 m diameter LFL methane spherical gas cloud for areas where criticality/congestion/gas volume are high, and ignition potential is med to high and 7 m for all other areas requiring gas detection.
- **Engineering Calculation**: For conservative and align with internal design specification, the mass of gas within spherical clouds of 5m and 7m diameter is calculated to estimate the quantity of gas release for congested area and well-ventilated area, respectively.

The mass integration in spherical coordinates is well-established principles and widely used and taught in fluid mechanics, chemical engineering, and atmospheric dispersion modeling. The integration of a concentration profile over a spherical volume to determine gas mass is based on standard spherical coordinate volume integration, as presented in transport phenomena and process safety literature (e.g., Bird et al., 2002; Crowl & Louvar, 2011; Lees, 2012). The total mass of a gas distributed in a spherical volume with a varying concentration profile:

$$m = \rho_{gas}\int_0^R\int_0^{2\pi}\int_0^{\pi}\left(\frac{C(r)}{100}\right)r^2\sin\theta \, d\theta \, d\emptyset \, dr \qquad \text{Equation 1}$$

Where:

- *Pgas* is the gas density (e.g., methane),
- *C* (r) is the concentration (%) at radius r,
- $\left(\frac{C(r)}{100}\right)$ gives the volume fraction of gas,
- $r^2 \sin\theta$ is the Jacobian (volume element) in spherical coordinates.

Based on the calculations for oil and gas facilities where methane is the primary contained gas, a pre-determined release threshold is adopted to simplify assessments for frontline personnel. The minimum mass of a release with the potential for high consequence is conservatively estimated at 5.4 kg in congested areas and at 10.6 kg in well-ventilated areas. These thresholds serve as criteria to assess whether the quantity of released gas is sufficient to present a credible risk of fire or explosion.

*Liquid Release Evaluation*

In the context of Loss of Primary Containment (LOPC) involving liquid releases, it is essential to assess the potential for fire escalation based on the adequacy of containment infrastructure. This evaluation focuses on the design and effectiveness of secondary containment systems such as curbs, bunds, and trays. Properly engineered containment systems are capable of localizing spills, thereby significantly reducing the risk of fire propagation. Conversely, the absence of such containment, or scenarios where the containment capacity is exceeded, can lead to the uncontrolled spread of flammable liquids. This increases the likelihood of a non-localized fire event, posing a greater threat to personnel, equipment, and the environment. Therefore, the presence, design integrity, and capacity of secondary containment systems must be critically examined when classifying LOPC events.

Liquid releases are assessed based on containment design:

- Properly designed curbs, bunds, or trays can localize spills and reduce fire risk.
- Absence or overflow of secondary containment increases the likelihood of widespread fire scenarios.

This step ensures that the potential for escalation is not underestimated and that incidents with inadequate containment are appropriately flagged as High Potential Incidents (HPIs). Incorporating containment evaluation into the classification process enhances the accuracy of risk assessments and supports more effective mitigation planning.

***Verification of Mitigation Safeguards.*** This step involves evaluating whether adequate mitigation barriers are in place to prevent the escalation of a hydrocarbon gas or liquid release—classified under B.1—into a Major Accident Event (MAE). Among the most critical mitigation measures are the control of ignition sources and the deployment of fire and gas detection systems. These barriers play a pivotal role in reducing the likelihood of fire and explosion by enabling early detection and intervention, thereby enhancing the overall integrity of the safety management system.

Control of Ignition Sources

The control of ignition sources is a fundamental principle in process safety management, particularly in environments where flammable substances are handled or stored. The theoretical basis for controlling ignition sources lies in the fire triangle, which identifies three essential elements for combustion: fuel, oxygen, and an ignition source. In industrial settings, while the presence of fuel and oxygen may be unavoidable, the ignition source is the most controllable variable. Therefore, systematic identification, elimination, or mitigation of ignition sources is critical to preventing fire and explosion incidents.

Effective management of ignition sources not only reduces the likelihood of incident escalation but also enhances the overall integrity of process safety systems. It is a key component in evaluating the risk profile of Loss of Primary Containment (LOPC) events and determining their classification as High Potential Incidents (HPIs).

Fire and Gas Detection Systems

Fire and Gas Detection Systems (FGDS) are critical components of industrial safety infrastructure, designed to provide early warning of hazardous conditions that may lead to fires, explosions, or toxic exposures. These systems operate on the principle of continuous monitoring and rapid response, enabling timely intervention to prevent escalation of incidents.

Upon detection, FGDS can trigger alarms, activate emergency shutdown systems, initiate fire suppression, and alert personnel. Their effectiveness depends on proper design, calibration, maintenance, and integration with other safety systems.

In the context of LOPC-related HPI classification, the presence and performance of FGDS are crucial in determining whether a release has credible escalation potential. A confirmed detection event may serve as evidence of an effective barrier, while absence or failure of detection may elevate the incident's risk profile.

In Part B, these criteria reflect a more conservative and precautionary approach, emphasizing the importance of thorough evaluation in scenarios where immediate classification may not apply.

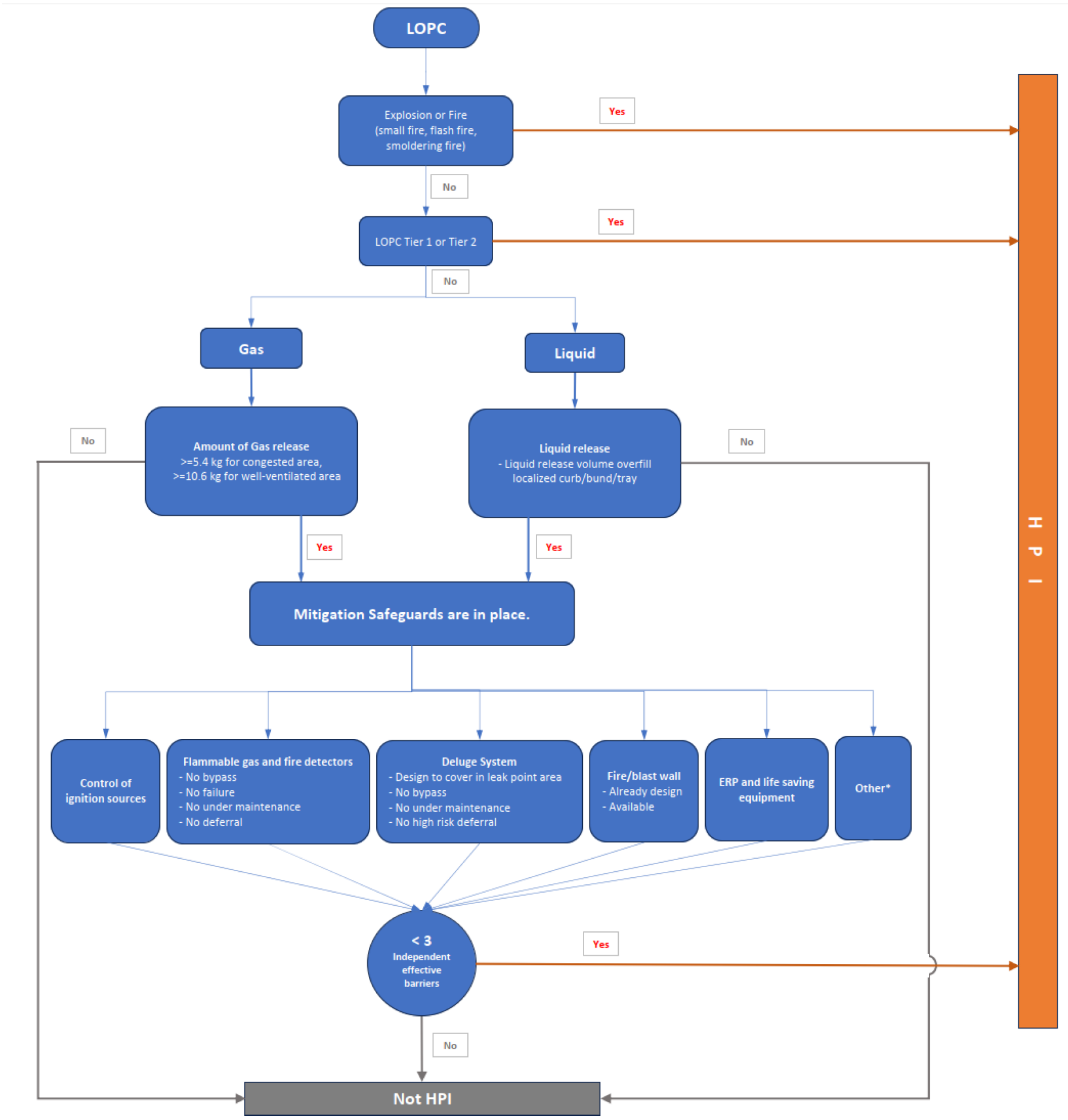


Figure 1—Analytical Process for Determining High Potential Incidents (HPI)

## Data and Results

To ensure consistency and ease of use in the field, the proposed HPI classification methodology has been distilled into a simplified checklist, as presented in Appendix A. This checklist integrates all elements from both the Immediate HPI Classification and Conditional HPI Classification processes. Designed with operational practicality in mind, the checklist enables field personnel to apply the criteria efficiently and reliably across diverse scenarios. Its user-friendly format supports rapid decision-making while maintaining the rigor of the underlying methodology.

### Integration with IMS

To facilitate seamless application across operational environments, the HPI classification criteria have been embedded within the Incident Management System (IMS). This integration ensures that the methodology is systematically applied during incident reporting and investigation, enhancing consistency and traceability. Embedding the criteria into IMS also supports automated workflows, enabling real-time decision support and alignment with broader safety management protocols.

**Pilot Study Overview**

To validate the effectiveness and applicability of the newly developed HPI criteria for LOPC events, a pilot study was conducted using 25 real incident cases from various operational assets, both domestic and international. The primary objective of this study was to assess whether the criteria could consistently and accurately identify incidents with credible potential for escalation into Major Accident Events (MAEs).

**Key Observations**

- Consistency: The criteria demonstrated a high degree of consistency in evaluating a wide range of incident types. Regardless of the asset location or operational context, the methodology provided a uniform framework for classification.
- Clarity: The use of structured flowcharts and quantitative thresholds significantly reduced ambiguity in decision-making. Field personnel were able to apply the criteria with confidence, minimizing subjective interpretation.
- Barrier Effectiveness: The requirement for verifying the presence of mitigation barriers—such as fire and gas detection systems and ignition source controls—proved instrumental in distinguishing between high and low potential incidents. This aspect of the methodology reinforced the importance of safeguard integrity in incident evaluation.

**Case Classification Summary**

Out of the 25 cases reviewed:

- 3 incidents were classified as High Potential Incidents (HPIs), including:
  - A fire incident
  - A Tier 2 LOPC event
  - A confirmed gas release with insufficient mitigation barriers
- 22 incidents were not classified as HPIs, based on adequate containment, limited release volume, or effective barrier implementation.

These results underscore the robustness of the criteria in handling a variety of scenarios and highlight the importance of accurate data and documentation in incident classification. The pilot study also demonstrated that the methodology is scalable and adaptable to different operational environments, supporting its broader implementation across the organization.

**Challenges and Continual Improvement**

During the development of the methodology, the primary challenge was identifying the optimal balance—or "sweet spot"— among three key factors: ease of use for front-line operators, accuracy in identifying High Potential Incident (HPI) cases (i.e., ensuring no HPI case is missed), and avoiding the over-classification of minor incidents.

To address this, a simplified, pre-determined minimum hydrocarbon release mass threshold was introduced. This approach enables operators with minimal training to apply the criteria effectively using a checklist, and it is unlikely to overlook any HPI case. However, it may conservatively classify a greater number of minor cases as HPIs.

Alternatively, a more rigorous approach involves simulated calculations to determine whether a release results in a significant gas cloud—defined as 5 meters in enclosed areas and 7 meters in open areas. While this method could enhance classification precision, it is time-consuming and requires personnel with engineering expertise, which may hinder timely identification.

The implementation of this methodology is being closely monitored through Safety Statistics at both the Asset and Enterprise levels. Adjustments to the simplified criteria may be considered as needed, based on ongoing performance and feedback.

## Conclusions

The development and implementation of standardized HPI criteria for LOPC events represent a significant advancement in process safety management. By integrating engineering principles, empirical data, and barrier-based safety models, the methodology offers a clear, objective, and replicable framework for identifying incidents with credible escalation potential. This approach enhances the classification of process safety events and supports the proactive prevention of LOPC-related scenarios.

Through its structured design, the methodology elevates the level of incident investigation, strengthens the effectiveness of corrective actions, and enables systematic extraction of lessons learned from events with high potential consequences. These improvements directly contribute to the overarching goal of achieving zero incidents across operational environments.

### Key Achievements

- Reduced Subjectivity: The criteria minimize reliance on individual judgment, leading to more consistent and defensible classifications.
- Enhanced Safety Awareness: Accurate identification of HPIs enables organizations to prioritize preventive actions and allocate resources more effectively.
- Scalability: The framework is applicable across various operational contexts, including offshore platforms, onshore facilities, and international assets.

## Reference

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# Appendix

## Appendix A Simplified LOPC-Related HPI Checklist